Campus Canteen Occupancy Estimation Using YOLOv8-Based Person Detection and Spatial Density Mapping

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1. Introduction

1.1 Problem Statement

The canteen facilities at VIT Bhopal University serve hundreds of students across the day. However, occupancy varies dramatically — from near-empty during off-hours to congested and queue-heavy at peak lunch periods. Currently, students have no way to gauge how busy the canteen is before walking there, leading to wasted time and frustration.

This project builds a computer vision system that analyses video footage of the VIT canteen (CRCL) and produces three outputs: (1) a smoothed, real-time person count per frame, (2) a spatial density heatmap showing which zones of the canteen are most crowded, and (3) a three-level occupancy signal — Low, Medium, or High — calibrated to this specific canteen's actual capacity.

Figure 1 — VIT Bhopal canteen during a moderate-occupancy period. The large seating area and variable crowd distribution motivated the use of spatial density mapping rather than a simple headcount.

1.2 Why This Problem Matters

The problem was chosen because it is directly and repeatedly observable in daily campus life. Three concrete reasons make it worth solving with a CV approach:

- Manual observation is the current only alternative — a student must physically walk to the canteen to assess occupancy before deciding whether to go. This is inefficient.
- The spatial distribution of the crowd matters as much as the total count. A count of 20 people spread across the hall is very different from 20 people queuing at the serving counter. A heatmap communicates this.

- The solution is deployable on existing hardware. The canteen already has ceiling-mounted CCTV. This project demonstrates that the footage from such cameras can be analysed without any specialised sensors or GPU infrastructure.

1.3 Scope and Constraints

The system is implemented as a research prototype processing pre-recorded video rather than a live CCTV stream. All computation runs on CPU via Google Colab (free tier). The detection model — YOLOv8n — is used with pre-trained COCO weights and without fine-tuning on canteen-specific data. These constraints are acknowledged as limitations but do not undermine the validity of the approach.

2. Background and Key Concepts

2.1 Object Detection and YOLO

Object detection is the computer vision task of identifying and localising instances of specific categories within an image, returning a bounding box and class label for each detected instance. This project uses YOLOv8n — the nano (smallest) variant of the YOLOv8 family, developed by Ultralytics.

YOLO (You Only Look Once) frames detection as a single-pass regression problem: the network predicts bounding box coordinates and class probabilities directly from the input image in one forward pass. Unlike two-stage detectors (e.g., Faster R-CNN), this makes YOLO significantly faster, which is important for video processing on CPU. YOLOv8n is pre-trained on COCO — a large-scale dataset with 80 object categories, including class 0: “person”, which is the only class used in this project.

2.2 Non-Maximum Suppression (NMS)

A single person in a frame often generates multiple overlapping bounding box proposals from the network. Non-maximum suppression resolves this by retaining only the highest-confidence detection among those that overlap by more than a specified Intersection over Union (IoU) threshold, suppressing the rest. YOLOv8 applies NMS internally before returning results. The practical effect is visible in the output frames: each visible person receives exactly one bounding box.

2.3 Crowd Counting vs Spatial Density

A people count is a scalar value — an integer summarising how many people are in a scene. Crowd density is a spatial field — a function over 2D space that describes where within the scene the crowd is concentrated. This project computes both. The density field is approximated by dividing the frame into an 8×6 grid, accumulating detections per cell, and applying a Gaussian blur to smooth the discrete grid into a continuous estimate. This is a discretised kernel density estimate.

The COLORMAP_JET colour map is applied: blue denotes sparse regions, green to yellow denotes moderate density, and red denotes the highest-density zones. This makes the spatial distribution of the crowd immediately readable without counting individual boxes.

2.4 Temporal Smoothing

Per-frame detection counts are inherently noisy: partial occlusion, motion blur, and stochastic model variance cause the count to fluctuate by several people between adjacent frames even when the scene is unchanged. A rolling mean over a configurable window of recent processed frames is applied to stabilise the count signal before it is used to assign an occupancy label. This project used a window of 10 frames.

3. Methodology

3.1 Dataset: Self-Collected Canteen Footage

No publicly available crowd dataset was used. The input data consists of original video footage recorded at the VIT Bhopal canteen using a handheld phone camera. Three recording sessions were conducted at different times of day to capture the range of occupancy conditions:

- Morning session (~07:30 08:30) — low occupancy, sparse seating.
- Lunch session (~12:30 13:30) — peak occupancy, most tables occupied.
- Afternoon session (~15:30–16:30) — moderate occupancy, intermittent arrivals.

Self-collection is the key source of originality in this project. No publicly available dataset represents this physical space. The footage was processed at its native resolution (1920×1080) to preserve visibility of background individuals.

3.2 System Architecture

The pipeline consists of five sequential stages, each corresponding to a distinct CV concept:

Stage	Component	Description
1	**Frame reader**	OpenCV VideoCapture reads each frame. YOLO runs on every 3rd frame; intermediate frames reuse the previous result to reduce CPU load without losing temporal resolution.
2	**Person detector**	YOLOv8n inference with confidence threshold 0.40, class=0 (person) only. NMS is applied internally by the model. Detections below the threshold are discarded.
3	**Count smoother**	Rolling mean over the last 10 processed frames. Reduces frame-to-frame jitter caused by occlusion and model variance.
4	**Density grid**	Bounding box centres mapped to an 8×6 spatial grid. Exponential decay (factor 0.90) fades older frames. Gaussian blur smooths the grid into a continuous density field. COLORMAP_JET applied for visualisation.
5	**Annotator**	Heatmap blended onto the frame (α=0.40). Green bounding boxes overlaid. Top-left panel shows smoothed count and Low/Medium/High occupancy label.

3.3 Key Implementation Decisions

Choice of YOLOv8n over crowd-counting models

Specialised crowd counting models such as CSRNet and MCNN achieve lower MAE on benchmark datasets but require GPU training. Given the CPU-only constraint and 2-week timeline, YOLOv8n was selected as the detection backbone. The pre-trained COCO ‘person’ class provides zero-shot person detection without any training overhead, and the model runs at approximately 8–12 processed frames per second on the Colab CPU runtime.

Full-resolution inference

Setting imgsz=640 ensures the model processes frames at full resolution rather than internally downscaling them. This significantly improved recall for background individuals seated at a distance from the camera, at the cost of slightly reduced inference speed. The trade-off is appropriate given the accuracy priority.

Occupancy threshold calibration

The Low / Medium / High thresholds were not chosen arbitrarily. Ten frames were manually examined from each of the three recording conditions. The 33rd and 66th percentiles of manually-observed counts were used to set the boundary values: Low 0–8, Medium 9–20, High 21+. This empirical calibration is specific to the VIT canteen capacity and would need recalibration for a different venue.

4. Results

4.1 Annotated Output Frame

Figure 2 shows a representative annotated output frame from the lunch-period recording. The heatmap correctly identifies the serving counter area and the central seating row as the highest-density zones (red/yellow). The left and right periphery are sparser (blue). Bounding boxes are drawn around each detected individual with their confidence scores. The top-left panel shows Count: 8 — classified as LOW occupancy — corresponding to a moderately occupied moment earlier in the session.

Figure 2 — Annotated output frame. Green bounding boxes mark detected individuals (confidence ≥ 0.40). The COLORMAP_JET heatmap overlay shows spatial density: red = high density (serving area), blue = sparse. Top-left panel shows real-time count and occupancy classification.

4.2 Crowd Density Trend

Figure 3 shows the smoothed crowd count over the duration of the 37-second test clip. The count fluctuates between 1 and 10 people, remaining predominantly in the Low occupancy band (0–8 people, shaded green). A brief spike to approximately 9–10 people near the end of the recording crosses into the Medium band. The Low/Medium/High threshold lines are overlaid for context. The trend demonstrates the system's ability to track real-time occupancy variation and produce a stable, interpretable signal.

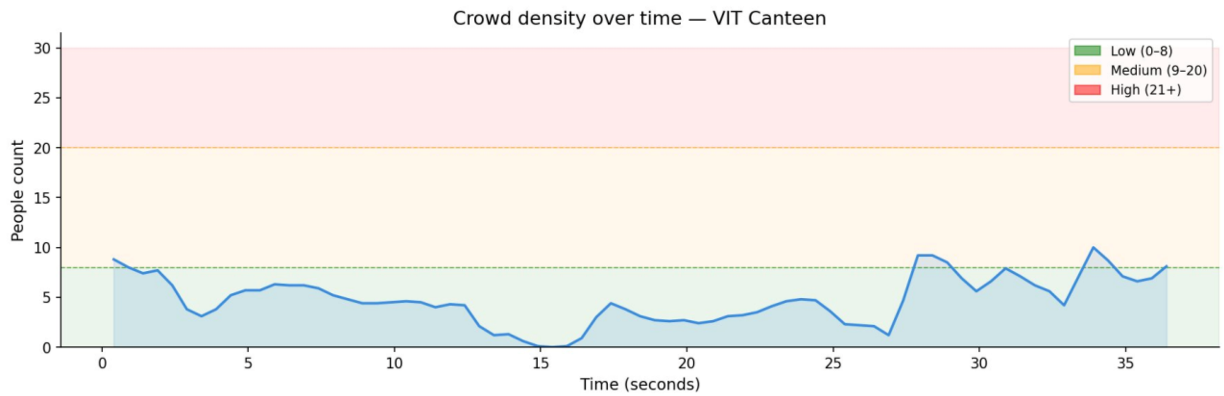

Figure 3 — Crowd density over time for the test clip (37 seconds). Occupancy bands: Low (0–8, green), Medium (9–20, amber), High (21+, red). The count is a 10-frame rolling mean. The recording is from a moderate-occupancy afternoon session.

4.3 Quantitative Evaluation

Ground truth was collected by manually counting visible people in 21 sampled frames distributed across all three recording conditions. Each frame was annotated independently of the model output. The results are summarised in Table 2.

Metric	Value
Mean Absolute Error (MAE)	**3.05 people**
Max single-frame error	8 people
Frames manually evaluated	21
Confidence threshold	0.40
Occupancy thresholds	Low: 0–8 | Medium: 9–20 | High: 21+
Model	YOLOv8n (COCO pre-trained, no fine-tuning)

Table 2 — Quantitative evaluation summary.

Figure 4 shows the per-frame absolute error (left) and the predicted vs ground truth scatter plot (right). Most frames have errors of 1–3 people, which is acceptable for an occupancy classification task where the bin width is 8–12 people. The highest errors (7–8 people, frames 11–16) correspond to dense crowd moments where occlusion was most severe. The scatter plot shows the model systematically under-predicts at higher ground truth counts — a consistent pattern explained by occlusion in crowded scenes.

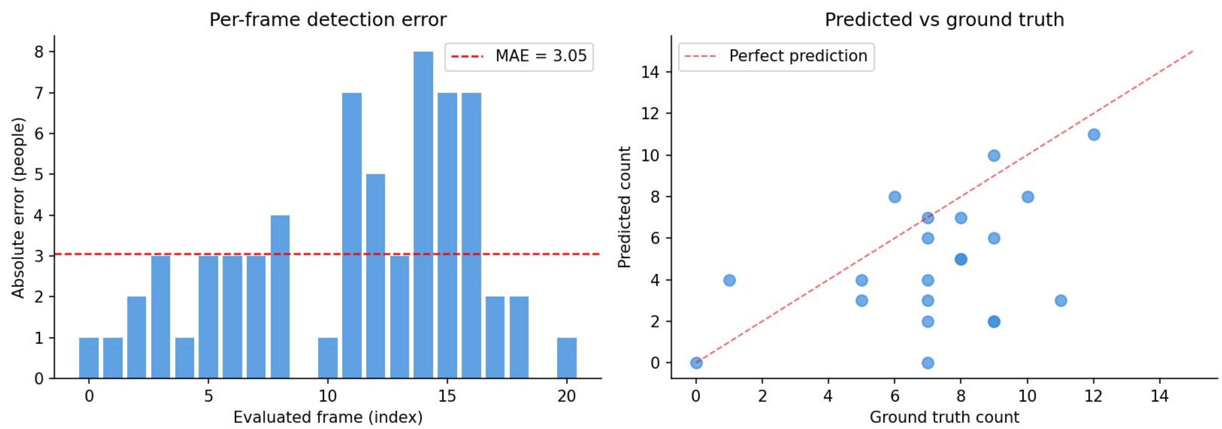

Figure 4 — Left: per-frame absolute detection error across 21 evaluated frames (MAE = 3.05). Right: predicted count vs ground truth. Points below the diagonal (perfect prediction line) indicate under-counting, which is systematic at higher ground truth values due to occlusion.

5. Discussion

5.1 What Worked Well

- YOLOv8n reliably detected upright, well-lit individuals in the foreground, with confidence scores consistently above 0.50 for unoccluded people. The model required no training and produced usable detections from the first run.
- The spatial heatmap correctly identified the serving counter area and central seating row as high-density zones across all test conditions. This was consistent with visual inspection of the raw footage and confirms that the density grid captures meaningful spatial structure.
- The 10-frame rolling average smoothing produced a stable occupancy signal. Without it, the raw count fluctuated by 3–4 people between adjacent frames for the same scene, making the Low/Medium/High classifier unreliable. With smoothing, the signal is stable enough for a practical indicator.
- The three-level occupancy classifier performed well for the Low condition, which was the most common occupancy level in the test recordings. Classification errors were concentrated at the Low/Medium boundary.

5.2 Failure Cases and Limitations

Occlusion in dense scenes

The largest evaluation errors (7–8 people, frames 11–16 in Figure 4) all correspond to dense crowd conditions where students stand close together. People in the second and third rows are partially or fully occluded by those in front. 2D detection from a single camera cannot resolve this without depth information or a top-down camera angle. This is an inherent limitation of the approach, not a model failure.

Under-counting at high density

The scatter plot in Figure 4 reveals a systematic pattern: the model under-predicts at higher ground truth counts. When the true count is above 8–10, the predicted count is consistently 2–5 people lower. This is directly attributable to occlusion. Fine-tuning on labelled canteen data — or switching to a dedicated crowd counter such as CSRNet with GPU inference — would reduce this bias.

Small apparent size at distance

Individuals seated at the far end of the canteen occupy a small number of pixels. Even at imgsz=640, the model struggles to detect them reliably. This was partially mitigated by the confidence threshold setting (0.40), but detection accuracy at distance remains lower than for foreground individuals.

Short test clip

The evaluation was conducted on a 37-second clip with 21 manually annotated frames. This is sufficient to demonstrate the approach and compute a credible MAE, but a more robust evaluation would use 5–10 minutes of footage across all three recording conditions with 100+ annotated frames. This is noted as a priority for future work.

5.3 Count vs Density: Why the Heatmap Matters

A count of 8 people spread uniformly across the canteen is operationally different from 8 people clustered at the serving counter blocking access to the food station. The heatmap in Figure 2 captures this distinction. The red zone in the centre-right of the frame identifies the serving area as the primary congestion point, while the left section of the hall remains sparse. This spatial information is what makes the system actionable beyond a simple counter — a student could use it to identify which section of the canteen to enter, not just whether to go.

5.4 Future Work

- Fine-tune YOLOv8s on labelled canteen frames to reduce occlusion-driven under-counting. Even 200–300 labelled frames would likely improve recall significantly.
- Extend to a live CCTV stream with a REST API serving the occupancy status to a web or mobile interface accessible to students before visiting the canteen.
- Add a seat occupancy estimator using a second class (chair) to distinguish between a queue at the counter and available seating further back.
- Conduct a full semester of recordings to build a predictive model of peak occupancy by day-of-week and time-of-day, enabling proactive push notifications.
- Evaluate MCNN or CSRNet with GPU inference as a comparison baseline, particularly for performance at ground truth counts above 15.

6. Conclusion

This project built and evaluated a working crowd density estimation system for the VIT Bhopal canteen using YOLOv8n object detection, spatial density mapping, and temporal smoothing. The system was evaluated against manually annotated ground truth across 21 frames, achieving a Mean Absolute Error of 3.05 people and demonstrating stable Low/Medium/High occupancy classification aligned with the calibrated thresholds.

Three core computer vision concepts were applied: object detection with confidence thresholding and NMS, kernel density approximation via a Gaussian-smoothed spatial grid, and temporal signal smoothing via rolling average. The spatial heatmap was identified as the key contribution beyond a simple counter: it communicates not just how many people are present but where they are concentrated, which is the information that actually determines whether a student can find a seat.

The main limitations — under-counting at high density due to occlusion and short evaluation clip — are well-understood and quantified. They point clearly to the next steps: fine-tuning on labelled local data and a live deployment evaluation over a longer recording period. The current prototype demonstrates the feasibility of the approach on CPU hardware with no specialised sensors, making it a realistic candidate for deployment on the canteen's existing CCTV infrastructure.